Exercises from previous weeks

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Outline

The goals of today's tutorial is detailed solutions to exercised that we haven't yet discussed from Weeks 2-4:

- Carry out the matrix chain rule homework from W3 Lecture 1 (slide 21)
- Detailed solution of Question 4 from W2 exercise sheet on Decision Trees
- Questions 3 and 4 from Week 4 Exercise sheet on generalisation and confidence bounds

Homework left from W3 Lecture 1 (slide 21): Matrix Chain Rule Derivation of $\nabla_w(w^\top X^\top X w)$

We want to compute the gradient:

$$f(w) = w^\top X^\top X w$$

using the matrix form of chain rule (Jacobian-based chain rule).

We will define intermediate functions:

$$G(w) = Xw, \qquad F(u) = u^\top u,$$

so that:

$$f(w) = (F \circ G)(w) = F(G(w)).$$

Step 1: Define the intermediate functions

Definitions of intermediate functions

$$G : \mathbb{R}^d \to \mathbb{R}^n, \quad G(w) = Xw,$$

$$F : \mathbb{R}^n \to \mathbb{R}, \quad F(u) = u^\top u.$$

The Jacobians of these functions

$$J_G(w) = \frac{\partial G}{\partial w} = X \in \mathbb{R}^{n \times d},$$

$$J_F(u) = \frac{\partial F}{\partial u} = 2u^\top \in \mathbb{R}^{1 \times n}.$$

Step 2: Applying the Chain Rule

Applying the matrix form of chain rule

$$\frac{\partial f}{\partial w} = J_F(G(w))\, J_G(w) = (2u^\top)X, \quad \text{where } u = G(w) = Xw.$$

Formatting the result to column gradient

The (column) gradient is the transpose of the row derivative:

$$\nabla_w f = \left(\frac{\partial f}{\partial w}\right)^\top = (2u^\top X)^\top = 2X^\top u = 2X^\top Xw.$$

Question 4 from W2 Exercise Sheet: Decision trees

Dataset (8 rows, categorical features):

Age Group	Income Level	Prior Purchase	Purchase (Y)
Young	Low	No	No
Young	Low	Yes	No
Young	High	No	Yes
Middle-aged	Low	No	No
Middle-aged	High	No	Yes
Senior	Low	No	No
Senior	High	Yes	Yes
Senior	High	No	Yes

We'll compute: entropy of Y, information gain of splitting on "Prior Purchase", best root attribute, and first split.

Q4(a) Compute the Entropy of the target Y

Count outcomes:

$$\#\text{Yes} = 4, \qquad \#\text{No} = 4, \qquad n = 8.$$

So $p_+ = p_- = 0.5$ and

$$H(Y) = -0.5 \log_2 0.5 - 0.5 \log_2 0.5 = 1.0 \text{ bit.}$$

(Highest possible entropy for a Bernoulli variable.)

Q4(b) Compute Information Gain for 'Prior Purchase'

Split by `Prior Purchase` (Yes / No):

- Prior Purchase = Yes: rows 2 and 7
  - Targets: {No, Yes} $\Rightarrow$ counts (Yes=1, No=1) so $p_+ = 0.5$ and

$$H(\text{Yes-child}) = H(Y \mid \text{`}Prior\ purchase' = Yes)$$
$$= -0.5 \log_2 0.5 - 0.5 \log_2 0.5 = 1.0.$$

- Prior Purchase = No: remaining 6 rows have targets (#Yes=3, #No=3) so again $p_+ = 0.5$ and

$$H(\text{No-child}) = H(Y \mid \text{`}Prior\ purchase' = No) = 1.0.$$

Weighted sum of entropies from the branches – that is, the conditional entropy $H(Y|feature) = H(Y|\text{``Prior purchase})$:

$$\frac{2}{8} \cdot 1.0 + \frac{6}{8} \cdot 1.0 = 1.0.$$

$\Rightarrow$ Information Gain:

$$IG(Y, \text{`}Prior\ Purchase') = H(Y) - 1.0 = 1.0 - 1.0 = 0.$$

Q4(c) Compute Information Gain for remaining features

Income Level (Low / High):

- Income = Low: rows 1,2,4,6 (targets = {No,No,No,No}) $\Rightarrow$ (Yes=0, No=4) so

$$H(Y \mid \text{'Income level'} = \text{Low}) = 0.$$

- Income = High: rows 3,5,7,8 (targets = {Yes,Yes,Yes,Yes}) $\Rightarrow$ (Yes=4, No=0) so

$$H(Y \mid \text{'Income level'} = \text{High}) = 0.$$

- $H(Y \mid \text{'Income level'}) : \frac{4}{8} \cdot 0 + \frac{4}{8} \cdot 0 = 0.$
- Information gain:

$$IG(Y, \text{'Income level'}) = H(Y) - 0 = 1.0 \text{ bit}.$$

Age Group (Young / Middle-aged / Senior):

- Young: rows 1,2,3 with targets {No,No,Yes} $\Rightarrow$ Yes=1, No=2 $\Rightarrow p_+ = 1/3$.

$$H(Y \mid \text{`Age group'} = \text{Young}) = -\tfrac{1}{3}\log_2\tfrac{1}{3} - \tfrac{2}{3}\log_2\tfrac{2}{3} \approx 0.9183.$$

- Middle-aged: rows 4,5 with targets {No,Yes} $\Rightarrow p_+ = 1/2$.

$$H(Y \mid \text{`Age group'} = \text{Mid}) = 1.0.$$

- Senior: rows 6,7,8 with targets {No,Yes,Yes} $\Rightarrow p_+ = 2/3$.

$$H(Y \mid \text{`Age group'} = \text{Senior}) \approx 0.9183.$$

- Conditional entropy for 'Age group' (weighted child entropy):

$$H(Y \mid \text{`Age group'}) = \frac{3}{8}\cdot 0.9182958 + \frac{2}{8}\cdot 1.0 + \frac{3}{8}\cdot 0.9182958 \approx 0.9387219.$$

- Information gain:

$$IG(Y, \text{Age}) = H(Y) - 0.9387219 \approx 1.0 - 0.9387219 = 0.0612781 \text{ bits}.$$

Q4(d) First split and resulting tree

Root: Income Level

First split

- Income = Low $\Rightarrow$ **Predict = No** (all 4 examples are No)
- Income = High $\Rightarrow$ **Predict = Yes** (all 4 examples are Yes)

Because both children are pure, the tree stops here and the leaves are terminal.

Final tree (text form):

- If Income = Low $\Rightarrow$ Predict = No
- If Income = High $\Rightarrow$ Predict = Yes

Remarks and interpretation

- The dataset is perfectly separable by `Income Level`.
- `Prior Purchase` gives no information at the root (IG=0).
- `Age Group` gives a small IG (0.0613 bits) but is inferior to `Income Level`.

Q4(e) — Avoiding overfitting in decision trees

Techniques to improve generalization:

- **Pruning**: Post-prune branches that do not decrease validation error (cost-complexity pruning).
- **Limit complexity**: Maximum depth, minimum samples per leaf, minimum information gain threshold.
- **Ensemble methods**: Bagging (Random Forests), Boosting (AdaBoost, XGBoost) significantly reduce variance.
- **Cross-validation**: Use CV to choose tree hyperparameters and pruning level.
- **Feature engineering / regularization**: Reduce noisy variables and restrict splits.

Question 3 from Week 4 Exercise sheet

1. What could go wrong if a decision tree classifies all training points correctly?
2. How can confidence bounds help choose tree depth?

Solution

(1) Overfitting:

- A very deep tree perfectly fits training data but it may generalise poorly to new data.
- It captures the noise too rather than just the signal.

(2) Using Confidence Bounds:

- If enough labelled data is available, use **test set bounds**.
- Otherwise, you can use **Occam's razor bounds** on the training set, especially if you have good bets on the classifiers in the set of classifiers.
- These provide high-confidence bounds on the true generalisation error.
- Important in safety-critical applications (e.g. medicine, autonomous driving).

Question 4 from Week 4 Exercise Sheet

a) Answer the following:

(i) Is zero test error necessarily good?

(ii) Can the Occam's razor bound justify that it's a good classifier?

(iii) Given $n = 30$, 4 test errors, find 90% and 95% confidence upper bounds on true error.

b) Model tested on 1000 emails, 850 correct.

(i) Compute 95% confidence upper bound on true accuracy.

(ii) Interpret the meaning of confidence interval.

Solution (a) — Discussion and Bounds

(i) Zero test error can be misleading if the test set is small or unrepresentative.
(ii) Occam's razor bound helps justify small true error if:

- Training sample is i.i.d.
- Prior bets were made before observing data.

(iii) Test set bound:

$$c_{\mathcal{D}} \leq \hat{c}_S + \sqrt{\frac{\ln(1/\delta)}{2n}}$$

For 4 errors out of 30:

$$\text{90\% bound: } 0.1333 + \sqrt{\frac{\ln(10)}{60}} \approx 0.329$$

$$\text{95\% bound: } 0.1333 + \sqrt{\frac{\ln(20)}{60}} \approx 0.357$$

Solution (b) — Email Classification Example

Observed error $= 0.15$ on $n = 1000$ samples.
Using the same bound:

$$c_{\mathcal{D}} \leq 0.15 + \sqrt{\frac{\ln(20)}{2000}} \approx 0.189$$

Interpretation:

- With 95% confidence, true error ≤ 0.189.
- A 95% confidence interval means that across many random test samples, about 95% of such intervals would contain the true accuracy.

Plan for Friday

- Solutions of W5 exercise sheet on Bayes Networks + more exercises
- Q/A on any of the material - please post questions on Teams before Friday to have them discussed in the class!
- Approximate inference in Bayes networks (optional enrichment material)